

INTERNSHIP REPORT
OF
PROMPT ENGINEERING AND GENERATIVE AI

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B. Tech CSE



By

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Batch: 2CSE_14

2025-29

2026-27

CANDIDATE’S DECLARATION

I **Shashank Raj** do hereby declare that the internship report titled **“PROMPT ENGINEERING AND GENERATIVE AI”** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Name: Shashank Raj

Roll No: 25SCS1003001227

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature

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INTERNSHIP COMPLETION CERTIFICATE

4. PROJECT DESCRIPTION

4.1 Introduction

This internship focused on practical learning in prompt engineering and Generative AI. The work was organized around four supplied activities: Prompt Creation, Prompt Improvement, Prompt Role-Play, and Prompt Workflow. Together, these activities progressed from writing effective individual prompts to refining vague instructions, assigning professional roles, and connecting multiple prompts into a complete workflow. The submitted tasks show that AI output becomes more relevant, structured, and actionable when the user clearly specifies role, context, objective, audience, constraints, output format, examples, evaluation criteria, interaction rules, and verification.

4.2 Organization Profile

The internship work was conducted in the context of an educational, skill-development programme associated with InAmigos Foundation, with the submitted activities centered on practical AI literacy and prompt-engineering skills. The supplied task documents emphasize learning by experimentation: students create prompts, test them using an AI tool, observe differences in responses, compare approaches, and document their learning. The activities therefore functioned as structured exercises in applying Generative AI to education, planning, communication, decision-making, career preparation, and project organization. A formal organizational profile or statutory details were not included in the supplied task documents; accordingly, this report does not add unsupported institutional claims.

4.3 Problem Statement

A common challenge when using Generative AI is that a short or vague request may produce a response that is technically reasonable but poorly matched to the user's actual purpose. The internship addressed this problem by experimenting with increasingly structured prompts. The work examined how missing context, audience, constraints, format requirements, role assignment, evaluation criteria, and workflow connections can affect response quality.

4.4 Project Objectives

Understand how prompt wording affects the relevance, quality, structure, and usefulness of AI responses.

Learn to construct prompts using role, context, objective, audience, constraints, examples, and output format.

Compare basic prompts with improved prompts and identify the techniques responsible for better outputs.

Study how role assignment changes the tone, priorities, structure, and decision-making approach of an AI response.

Build a multi-step prompt workflow in which each stage supports the next stage.

Apply prompt engineering to practical activities such as study planning, learning, coding practice, decision-making, interviews, events, and project planning.

Develop responsible AI-use habits, including verification, avoiding blind copying, and following institutional policies.

4.5 Scope of the Project

The project scope covers the prompt-engineering experiments contained in the supplied Task 1, Task 2, Task 3, and Task 5 documents. Task 1 contains ten prompt-creation examples; Task 2 contains five prompt-improvement examples; Task 3 explores five professional roles; and Task 5 demonstrates a four-step interview-preparation workflow. The work focuses on prompt design and AI-assisted planning rather than development of a production AI model.

4.6 Technologies and Tools Used

Tool / Technique	Use in Internship
Generative AI / AI tool	Creation, testing, refinement, comparison, explanation, planning, and mock-interview activities.
Microsoft Word	Preparation and organization of task documents and final internship report.
Prompt-engineering techniques	Role assignment, context, constraints, output formats, examples, evaluation criteria, interaction, and verification.
Tables and structured formats	Comparison of responses, schedules, decision matrices, roadmaps, checklists, and workflow stages.

4.7 System Architecture

The overall learning architecture can be represented as a cycle: define the goal → design the prompt → generate the response → evaluate the response → refine the prompt → obtain the final output. The workflow below summarizes the architecture used across the internship tasks.

Stage	Description
Input	User goal, situation, audience, constraints, and required outcome
Prompt Design	Role + context + objective + constraints + format + evaluation/interaction rules

AI Processing	AI interprets instructions and generates a response
Evaluation	Check relevance, clarity, completeness, structure, and usefulness
Refinement	Add missing context or constraints and improve the prompt
Output	Actionable answer, plan, explanation, decision, interview practice, or workflow

4.8 Methodology

The internship followed an experiment-and-refine methodology. Each activity began with a practical goal, followed by prompt construction, AI testing, observation, comparison, and documentation. The progression from individual prompts to connected workflows helped convert prompt engineering from a theoretical concept into a repeatable practical skill.

Phase	Activity	Work Performed
Phase 1	Prompt Creation	Created 10 prompts for planning, learning, problem-solving, decision-making, summarization, idea generation, writing improvement, critical thinking, interview preparation, and goal-setting.
Phase 2	Prompt Improvement	Created five basic prompts, identified ambiguity, wrote improved versions, compared responses, and recorded the prompt-engineering technique used.
Phase 3	Role-Play Prompting	Tested Teacher, HR Professional, Event Planner, Career Advisor, and Project Manager roles and compared their priorities and response styles.
Phase 4	Prompt Workflow	Connected four prompts for a Software Developer

		interview: understand the role, generate questions, practice answers, and create a final seven-day preparation plan.
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Task 1 – Prompt Creation

Task 1 explored ten everyday and academic situations. The task demonstrated that specifying context, constraints, audience, format, role, evaluation criteria, and interaction rules can make AI responses more focused and useful. The task concluded that clearer instructions lead to better context and better AI responses.

No.	Purpose	Activity	Main Technique	Result
1	Planning	Study schedule	Context + constraints + format	Structured schedule
2	Learning	Embeddings in Generative AI	Role + simple language + analogy	Easy explanation
3	Problem-solving	Aptitude problem	Step-by-step + verification	Detailed solution
4	Decision-making	₹3,000 choice	Criteria + scoring	Balanced recommendation
5	Summarization	AI in everyday life	Word limit + structure	Concise summary
6	Idea generation	Technology projects	Constraints + ranking	Creative ideas
7	Writing improvement	AI-use paragraph	Audience + tone + word limit	Professional rewrite
8	Critical thinking	AI in academic assignments	Pros + cons + conclusion	Balanced analysis
9	Interview	Internship mock interview	Role + interaction rules	Interactive practice
10	Goal-setting	30-day Java plan	Time limit + stages	Actionable plan

Key learning: a well-designed prompt tells the AI what to do, who the response is for, what limitations to follow, and what format or outcome is expected.

Task 2 – Prompt Improvement

Task 2 compared five basic prompts with improved versions. The improved prompts added missing information such as learner level, available time, purpose, structure, scoring criteria, measurable tasks, and fallback instructions. The activity showed that prompt engineering is iterative: if the first answer is too general, the prompt can be refined by adding the missing information.

No.	Area	Original Prompt	Technique Used
1	Learning a concept	“Explain machine learning.”	Audience + structure + constraints + examples
2	Study planning	“Make a study plan for my exams.”	Context + time constraints + priorities + format
3	Summarization	“Summarize Generative AI.”	Purpose + word limit + required content + tone
4	Programming skills	“How can I improve my programming?”	Specific goal + progression + measurable tasks + fallback
5	Decision-making	“Which skill should I learn first?”	Context + evaluation criteria + scoring + recommendation

Overall finding: improved prompts were more immediately usable, more consistent, and better aligned with the intended purpose than the basic prompts.

Task 3 – Prompt Role-Play

Task 3 tested five professional perspectives: Teacher, HR Professional, Event Planner, Career Advisor, and Project Manager. Each role changed what the AI considered important and how it organized the response.

Role	Tone	Main Focus	Response Style
Teacher	Educational and supportive	Understanding concepts	Explanations, examples, questions

HR Professional	Professional and evaluative	Employability and behavior	Interview questions, sample answers, HR reasoning
Event Planner	Practical and organized	Execution and logistics	Schedule, resources, engagement, risks
Career Advisor	Motivational and goal-oriented	Long-term development	Roadmap, milestones, measurable outcomes
Project Manager	Structured and execution-focused	Delivery and accountability	Scope, roles, deadlines, dependencies, risks, testing

The Project Manager example was identified as the strongest example because the role naturally supports planning, task allocation, risk management, deadlines, testing, and accountability. The experiment demonstrated that role + context + clear instructions produces a more focused result than a broad request.

Examples of Role-Based Applications

Teacher: explaining Generative AI with simple language, examples, and understanding-check questions.

HR Professional: practicing internship interview questions and improving confidence, honesty, teamwork, and professionalism.

Event Planner: preparing the TechVision 2026 schedule, technical requirements, volunteer responsibilities, promotion, risk management, and event checklist.

Career Advisor: creating a six-month roadmap covering programming, data structures, AI fundamentals, projects, Git/GitHub, communication, resume, LinkedIn, interviews, and portfolio development.

Project Manager: planning a four-week AI Student Study Assistant with scope, team roles, milestones, dependencies, communication, risks, testing, success criteria, and presentation requirements.

Task 4 – Prompt Workflow

Task 5 demonstrated how several prompts can be connected to complete a larger practical task. The chosen activity was preparing for a Software Developer job interview. Each stage used the output or direction from the previous stage, creating a coherent progression from role understanding to final preparation.

Step	Activity	Purpose
Step 1	Understand the Job Role	Identify technical skills, soft skills, responsibilities, and preparation topics.
Step 2	Generate Interview Questions	Use Step 1 to create 8 technical, 4 HR/behavioural, and 3 problem-solving questions.
Step 3	Practice & Improve Answers	Ask questions one at a time, score answers on five criteria, and give specific improvement suggestions.
Step 4	Create Final Preparation Plan	Convert role analysis and feedback into a seven-day preparation plan and interview-day checklist.

The final seven-day preparation plan covered Java/C++ fundamentals, OOP, arrays/searching/sorting, SQL, HR preparation, a full mock interview, and final revision. The interview-day checklist included an updated resume, project review, programming revision, self-introduction practice, questions for the interviewer, professional dress, punctuality, confident body language, careful listening, and staying calm when an answer is unknown.

4.9 Expected Outcomes

Improved ability to write clear, specific, and goal-oriented prompts.

Better understanding of how audience, context, constraints, and output format influence AI responses.

Ability to compare a vague prompt with a structured prompt and explain why the output differs.

Understanding of role-based prompting and how professional perspectives change response priorities.

Ability to design connected prompt workflows instead of treating AI as only a one-question/one-answer tool.

Improved planning, communication, interview preparation, and project-organization skills.

Greater awareness of responsible AI use, verification, independent thinking, and academic integrity.

4.10 Certificates of completion and Other Supporting Documents

- Offer Letter
- Certificate Of Internship Completion
- Certificate of Best Intern
- Additional Certificate of Internship
- Certificate Of Appreciation

5. BIBLIOGRAPHY / REFERENCES

- 1. InAmigos Foundation. Internship-related certificates and official documentation provided to the student.**
- 2. Internship Report Format, School of Computer Science and Engineering, IILM University, Greater Noida, U.P.**
- 3. Internship learning materials, research notes, analytical observations, and AI-assisted work prepared during the PROMPT ENGINEERING AND GENERATIVE AI.**



DryRegistration Number –
U85300CT2020NPL010641 CSR - 1 Registered -
CSR00083159

22/08/2026

Offer Letter

Dear **Shashank Raj**,

We are pleased to offer you the internship position of **[AI PromptEngineering]** at InAmigos Foundation. Below are the details of your internship:

Duration of Internship: [Two Weeks].

Working Hours: The working hours for this internship are flexible, allowing you to adjust according to your availability. You are expected to dedicate a reasonable amount of time to the tasks and responsibilities assigned.

Location: This internship generally allows for remote work. However, if the role requires fieldwork, you will be expected to attend fieldwork at a site near your location.

Responsibilities

- Your primary role will be related to the field of your chosen internship. All major tasks assigned to you will align with this primary focus. Additionally, you will be expected to:
- Attend team meetings and actively participate in group discussions and activities.
- Complete tasks assigned by your superiors in a timely and efficient manner.
- Ensure that the tasks given to you are completed with the utmost care and responsibility.
- As a secondary responsibility, you will contribute to the organization's fundraising efforts, as this is crucial for your stipend.

Stipend: The stipend for this internship will be 20% of the donation raised by you during your internship period. The stipend is directly linked to the donations raised and may vary accordingly. Please note that while your main tasks will be aligned with your primary internship role, fundraising activities will be an essential part of your responsibilities to ensure the stipend. Fundraising is not a mandatory task for achieving a completion certificate. It is an optional task.

Terms and Conditions:

By accepting this internship, you agree to the following:

- InAmigos Foundation may use any content you provide, including reports, presentations, and other work, in a professional manner.
- Maintain professionalism and respect at all times.
- Adhere to all foundation policies and procedures during the internship.
- All work created during the internship will remain the property of InAmigos Foundation, unless otherwise agreed.
- Maintain confidentiality regarding sensitive information.
- Meet deadlines and complete tasks as assigned. Failure to do so may result in termination.
- Inform the foundation if you can no longer continue the internship, providing the appropriate notice.

Confidentiality: You are required to maintain the confidentiality of any information you receive during your internship. You are not permitted to disclose any sensitive or proprietary information related to InAmigos Foundation's operations or donors.

Termination: Either party may terminate this agreement with written notice, subject to the terms agreed upon.

Kindly sign below to confirm your acceptance of this offer and send the signed copy to foundation's email at hr@inamigosfoundation.org.in.

Authorized Signatory
Name: Ms. Shiwani
Designation: HR Head